

Acquisition Is Not Closure: Fail-Closed Authority for Open-World Scientific-Literature Discovery

Working framework draft

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Abstract

Scientific-literature discovery has two distinct control problems: acquiring useful evidence and deciding what that evidence is allowed to certify. Paper II separates an increasingly capable, donor-composable acquisition layer from a fail-closed scientific-closure authority layer. ORION counts routes as independent only when independence is earned through backend, query-derivation and capture identity; separates content identity from question-conditioned processing state; prevents route-local stopping from certifying task-global completeness; and retains unavailable or censored routes as explicit open obligations. In a frozen 390-task complete-gold controlled index, full ORION reaches mean recall 0.979487 versus 0.666667 for the strongest protocol-frozen confirmatory comparator and makes no premature task closures; targeted negative ablations expose the intended failure modes. The frozen statistical plan awards this campaign its committed `TIER_B.committed` precision tier, but the achieved half-width still exceeds the frozen superiority margin, so the primary offline result carries the plan’s mandatory underpowered label and these numbers support mechanism behavior rather than an inferential superiority claim. A separate credential-free MetaSyn probe, evaluated with the pinned official ID-only evaluator on all 86 released test reviews, reaches retrieval recall 0.7485 and inclusion recall 0.5651; it is a bounded external retrieval/screening probe rather than evidence of full multi-provider ORION superiority. A matched external ORION-vs-baseline Wide/Deep result is not archived in this revision, so the paper-level real-world superiority claim remains `CANNOT_CHECK`. The architectural contribution is a composable authority boundary beneath replaceable acquisition mechanisms: local utility, confidence, coverage, sufficiency or provider failure does not by itself erase unresolved scientific-search obligations.

Keywords: scientific literature discovery; information retrieval; deep research agents; systematic review; recall; stopping rules; open-world search.

1 Problem and claim boundary

Scientific discovery agents increasingly combine retrieval, query reformulation, reranking, graph expansion, selective reading and learned stopping. The central claim of Paper II is that progress in those components does not remove a separate systems question: *what authority does a local acquisition signal have to certify scientific completeness?* Paper II does not claim scientific RAG, agentic literature search, automated systematic review, capture–recapture estimation, lexical retrieval, reasoning-aware retrieval or learned stopping as novel. It studies the control boundary around those mechanisms: when is a route genuinely different, what has already been processed for the current scientific question, when may a route stop, and when must task-level coverage remain open?

The paper makes four scoped contributions. First, it formalizes *earned route independence*: tool or provider names carry no evidential weight without distinct backend, query-derivation and capture identities. Second, it separates route-local stopping from task-global closure and treats unavailable/censored routes as typed obligations rather than negative evidence. Third, it separates content identity from question-conditioned read state, preventing duplicate discovery inflation without suppressing legitimate rereading. Fourth, it makes the acquisition layer donor-composable: stronger lexical, dense, metadata-augmented, reasoning-aware, field-aware, graph-expansion, verification or utility mechanisms can be incorporated without inheriting authority to erase unresolved obligations.

Let route r be characterized by a backend identity b_r , query-derivation process q_r , and capture/content identity policy c_r . A distinct route label is insufficient for independence. Evidence for route independence must be derived from the actual generation and capture process rather than asserted as a boolean.

2 Question-conditioned cumulative read state

A paper being previously encountered is weaker than the statement that it has been processed for the current research question, extraction schema and content version. ORION therefore retains content identity separately from question-conditioned read state. A changed question or materially changed content can legitimately reopen a source for processing without pretending that it is newly discovered evidence.

3 Route stopping versus task stopping

Paper II separates route outcomes such as continue, reformulate, switch, budget-stop, unavailable and locally flat from task-level scientific closure. A route may stop because its marginal yield is low while another route remains untried, unavailable or censored. Coverage diagnostics therefore remain advisory; they cannot promote global completeness.

This fail-closed rule is especially important for capture-recapture-style diagnostics. Adaptive search routes are not automatically independent capture occasions, and zero overlap cannot justify a confident bounded estimate of unseen literature.

4 Acquisition is not closure

The central design move in Paper II is to separate two problems that are often collapsed into one policy. An *acquisition policy* decides what to search, retrieve, inspect, expand or reread next. A *closure authority* decides whether the evidence state is allowed to support route-local stopping or scientific task completion. Better acquisition can improve recall and reduce cost, but it does not automatically acquire authority to declare completeness. This separation makes strong external retrieval and stopping mechanisms composable rather than adversarial to the contribution of this paper.

Let x_t denote the observable search state at step t . We write an acquisition mechanism as

$$a_t \sim A(x_t),$$

where a_t may be a lexical query, a reasoning-aware query, a fielded Boolean filter, a graph-expansion action, a selective fetch, a reranking operation, or a request for another retrieval round. ORION

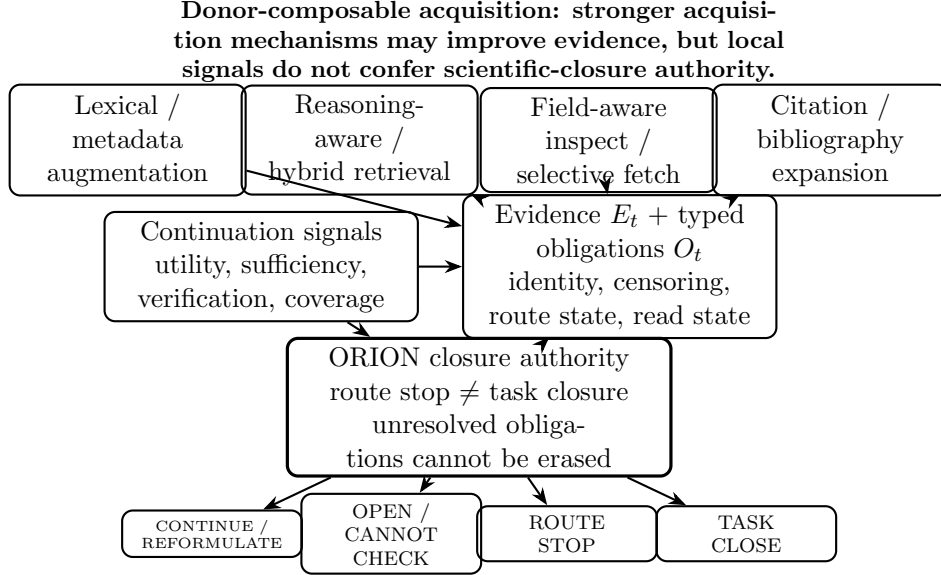


Figure 1: **Figure P2-7.** Acquisition mechanisms are intentionally composable: lexical/metadata augmentation, reasoning-aware or hybrid retrieval, field-aware selective fetching, citation expansion and continuation signals may all improve what is acquired. Their outputs update evidence and typed obligations, but a separate authority layer controls route stop, open or `CANNOT_CHECK` states and task closure. The figure therefore locates the candidate ORION residual at the meaning of acquisition outcomes rather than at a claim to have invented the strongest retriever.

places no novelty claim on the internal form of A . Instead, acquired evidence updates both an evidence state E_t and a typed obligation state O_t . Closure is a distinct map

$$C(E_t, O_t) \in \{\text{CONTINUE, ROUTE-STOP, OPEN, CANNOT-CHECK, TASK-CLOSE}\}.$$

A utility, relevance, confidence, sufficiency or coverage signal may influence A and may recommend a local stop, but it cannot by itself delete an unresolved obligation in O_t . This is the interface at which Paper II locates its candidate residual contribution.

4.1 Absorbing strong acquisition mechanisms

The acquisition layer is intentionally donor-composable. SAGE shows that keyword-oriented agent queries can make strong lexical retrieval difficult to beat and that corpus-side metadata/keyword augmentation can further improve retrieval [3]. AgentIR shows that an agent’s reasoning trace can be used jointly with its query as a retrieval signal [1]. SIEVE shows that fielded Boolean admission, structure-rich inspection and selective section fetching can improve the accuracy–context tradeoff [9]. Breadth-first bibliography expansion can recover literature missed by API-only retrieval, while also demonstrating why citation-list recall cannot be the only evaluation axis [8]. Paper II therefore treats lexical, metadata-augmented, reasoning-aware, field-aware and citation-expansion retrieval as candidate strengths to compose, not as ORION inventions.

The same rule applies to continuation and stopping signals. DeepControl uses marginal information utility to regulate continued retrieval [12]; decision-theoretic policies optimize screening stops against explicit costs and payoffs [2]; HALT uses claim–evidence coverage to verify whether expected evidence has been obtained [7]; MiCP applies conformal prediction to adaptive multi-turn

reasoning to maintain a target coverage guarantee while permitting early stopping [13]; and structured Search-R1 stopping judgments explicitly encode sufficiency and remaining gaps [4]. These signals and guarantees can all be useful inputs to acquisition or route-local stopping. The ORION-specific question is narrower: what are those signals permitted to certify when routes are dependent, unavailable, censored or only locally exhausted?

4.2 The authority residual

This decomposition yields four residual design claims that are separable from the mechanics above. First, route independence must be earned from backend, query-derivation and capture identity rather than from distinct tool names. Second, route-local stopping and task-global closure are different authority levels. Third, an unavailable or censored route remains a typed open obligation rather than becoming an empty result. Fourth, content identity and question-conditioned processing state must remain distinct so duplicate rediscovery does not inflate coverage while a genuinely changed scientific question can trigger justified rereading.

The resulting system is deliberately asymmetric. ORION is permissive about *how* evidence is acquired: a stronger retriever, query generator, expansion rule or verification signal can replace a weaker one. It is conservative about *what acquisition outcomes are allowed to mean*: local flatness, high estimated coverage, high confidence, apparent sufficiency or a provider failure cannot silently become proof of global completeness. This asymmetry is what permits the paper to absorb progress in retrieval without making its claimed authority boundary obsolete whenever a stronger retriever is published.

4.3 Empirical consequence

The decomposition also fixes the widening experiment. Candidate-generation methods should first be compared and composed under matched query/candidate budgets. Only after candidate recall is high enough to make stopping observable should the closure layer be challenged with direct ablations and strong stopping baselines. A positive external result must therefore show both acquisition quality and the incremental effect of typed closure authority; a cheaper stop that hides reachable relevant evidence is not a positive result. Conversely, a stronger donor retriever that improves acquisition while leaving ORION’s closure constraints intact strengthens the system rather than weakening the novelty claim.

5 Formal mechanics

This section fixes the objects the evaluation measures. Every definition below is implemented in the subject system rather than assumed: work identity and read state in `orion.knowledge.identity` and `orion.knowledge.ledger`, route independence and coverage in `orion.knowledge.routes`, and route control in `orion.knowledge.route_control`. Where a definition is deliberately conservative we state the failure it is conservative against, because the direction of the bias is what makes the refusal a safety property rather than a preference.

5.1 Work identity and content identity

Let R be the set of raw identifiers a route can emit (DOIs, arXiv ids, OpenAlex ids, local ids, URLs). Canonicalization is a map $\varphi : R \rightarrow \mathcal{S} \times V$ sending a raw identifier to a scheme-qualified key and a rendition version, so that $\varphi(\text{arXiv:2301.00001v1})$ and $\varphi(\text{arXiv:2301.00001v2})$ share

a key and differ only in V : a version is a rendition of one work, not a second work. A declared scheme is treated as a claim rather than a credential; an identifier that does not match the shape of the scheme it announces resolves to UNKNOWN instead of entering the graph as a key that can never be fetched.

A *work* is the alias-closure of canonical keys. Given identity records with alias sets $A_1, \dots, A_n$, works are the connected components of the graph on $\bigcup_i A_i$ that joins two keys sharing a record. Closure is computed as a projection over an append-only ledger rather than at write time, because an alias learned late can join two works recorded early; folding at write time would freeze the wrong partition.

Separately from work identity, each retrieved rendition carries a content digest c . Route overlap, deduplication and coverage are computed over content digests, never over retrieval identifiers: a source that re-mints identifiers would otherwise make two routes appear disjoint when they retrieved the same paper, which inflates every coverage quantity built on their overlap.

5.2 Question-conditioned read state

A read receipt is a tuple (w, c, σ, f) : work, content digest, extraction schema version, and frame f — the scientific question the extraction was performed for. Given the receipt set $\mathcal{R}$, the read decision for a candidate (w, c, σ, f) is

$$\text{read}(w, c, \sigma, f) = \begin{cases} \text{UNSEEN} & \nexists r \in \mathcal{R} : w_r = w \\ \text{CONTENT_CHANGED} & \exists r : w_r = w, \nexists r : (w_r, c_r) = (w, c) \\ \text{NEW_SCHEMA} & \exists r : (w_r, c_r) = (w, c), \nexists r : (w_r, c_r, \sigma_r) = (w, c, \sigma) \\ \text{NEW_FRAME} & \exists r : (w_r, c_r, \sigma_r) = (w, c, \sigma), \nexists r : r = (w, c, \sigma, f) \\ \text{ALREADY_READ} & \exists r \in \mathcal{R} : r = (w, c, \sigma, f). \end{cases}$$

ALREADY_READ therefore requires agreement on all four coordinates. This yields the distinction the evaluation depends on: a repeat extraction with all four coordinates equal is *duplicate processing* and is waste, while a repeat extraction under a changed content digest, schema or frame is a *legitimate reread* and is required work. Collapsing the two — treating “this paper has been seen” as “this paper has been read for this question” — is how a system reports coverage of material it only skimmed for a different purpose.

5.3 Earned route independence

A route capture is a tuple $\rho = (k, b, d, C)$: route kind, backend identity, query-derivation process, and the set C of captured content digests. For two captures ρ_i, ρ_j the independence predicate is

$$\text{ind}(\rho_i, \rho_j) \iff k_i \neq k_j \wedge b_i \neq b_j \wedge d_i \neq d_j,$$

and the verdict otherwise names the shared cause (SAME_ROUTE_KIND, SHARED_BACKEND, SHARED_QUERY_DERIVATION). Independence is thus *earned* from the generation and capture process; a distinct route label carries no evidential weight.

Proposition 1 (Direction of the conservatism). *Positive dependence between capture occasions biases unseen-mass estimators downward. Admitting a dependent pair therefore produces false confidence that search is near exhaustion, whereas refusing an independent pair only forfeits a diagnostic. The predicate is accordingly biased toward refusal.*

5.4 Coverage diagnostics are never authority

For an independent pair with captures of size n_i, n_j and overlap m_{ij} , the two-source estimate is $\hat{N} = n_i n_j / m_{ij}$, undefined at $m_{ij} = 0$. Zero overlap places no upper bound on the population, so the system returns a refusal with reason rather than a large number that would read as a measurement. When several independent pairs yield estimates, the reported value is $\max_{ij} \hat{N}_{ij}$: the largest population implies the most still unfound, which is the direction that fails safe. Every coverage estimate is marked diagnostic-only, and adaptive search is explicitly not a sequence of independent capture occasions — following up on what was just found converts singletons into doubletons and deflates the estimator toward “nearly done”.

5.5 Route control state machine

Let a route hold attempts $a_1, \dots, a_t$, each with retrieved digests D_i , novel digests $N_i \subseteq D_i$, cost, failure signatures and a backend-exhaustion flag. Marginal novelty of an attempt is $|N_t|/|D_t|$, and the flat streak z is the number of trailing attempts that succeeded, reported no exhaustion and returned no novel digest. Under a frozen policy with thresholds $\theta_{\text{ref}} \leq \theta_{\text{sw}}$, the decision is taken in strict priority order:

$$\delta = \begin{cases} \text{STOP (budget)} & \text{remaining budget} \leq 0 \\ \text{STOP (exhausted)} & a_t \text{ reports backend exhaustion} \\ \text{SUSPEND} & a_t \text{ carries execution-failure signatures} \\ \text{SWITCH} & z \geq \theta_{\text{sw}} \wedge \text{alternatives exist} \\ \text{REFORMULATE} & z \geq \theta_{\text{ref}} \\ \text{CONTINUE} & \text{otherwise.} \end{cases}$$

Two separations in this table are load-bearing. First, backend exhaustion and transport failure are distinct states: the former is a statement about the index, the latter about the channel, and a system that merges them converts an outage into evidence of absence. The latter yields SUSPEND, a non-terminal state carrying an open obligation, not STOP. Second, each terminal decision carries typed reopen triggers (budget increase, backend recovery or refresh, material frame change, material finding on an alternative route); a route reopens exactly when an observed trigger intersects that set.

Proposition 2 (Fail-closed stopping). *No route decision certifies task saturation. In the implementation the route decision exposes `certifies_task_saturation` $\equiv$ false for every action including STOP, so route-local exhaustion cannot be lifted to task-level completeness by any composition of route outcomes.*

Task closure is therefore a separate judgement over the route ensemble. It requires that every route is in a terminal or explicitly reopenable state, that no suspended or unavailable route leaves an unmet obligation, and — by Proposition 2 — that no coverage diagnostic has been used as its authority. An unavailable route remains an open coverage obligation and is reported as such; it is never converted into evidence that nothing further exists.

5.6 What the evaluation must therefore measure

The definitions above fix four measurable error families, each with a ground-truth condition that only a complete-gold world supplies: route-stop false positives (a route stopped while it could still

reach unfound relevant items), route-stop false negatives (a route continued past its exhaustion point), premature task closure (task closed while a gold item was still reachable within remaining budget), and duplicate processing versus legitimate reread as separated by the four read coordinates. Route overlap, per-route marginal contribution and earned-independence status are computed over content digests. Metrics whose denominator is the gold set are reported only where that denominator is complete; the conditions are stated per metric in the frozen measurement plan.

6 Methods

The design choices in this section were fixed before the result-bearing offline campaign. Post-outcome edits to this prose are limited to correcting execution status (for example, saying that an external benchmark has not yet been run) and do not change a hypothesis, task family, system, metric, budget, exclusion rule, margin or decision rule. Machine-readable artifacts govern: the design freeze in `protocol/PROTOCOL.V1.json`, measurement and statistical rules in `protocol/MEASUREMENT_PLAN.V1.md` and `protocol/STATISTICAL_PLAN.V1.json`, external-artifact states in `protocol/EXTERNAL_ACCESS_AUDIT.V1.json`, the controlled-world freeze under `evidence/offline_gold/`, and the prospective execution binding in `protocol/OFFLINE_RUN_MANIFEST.V1.json`.

6.1 Prospective discipline

The protocol is `P2.open-world-discovery.v1`. A result-bearing execution is admissible only after subject code, data, evaluator, system identities, budgets and repeat seeds are bound. For the completed offline companion, the run manifest records `outcome.accessed.before.freeze=false`, the exact subject revision, content hashes, 14 system identifiers, three repeat seeds and host-owned resource limits. The bound subject revision passed the full test suite before the first aggregate outcome snapshot was recorded. Publication summaries and manuscript prose may be generated after outcomes, but they may only project the frozen records; they cannot change the scientific decision rules.

6.2 Evaluation environments

Three environments are distinguished because no single one supplies both a complete denominator and open-world realism. Only the first has a completed result-bearing campaign in this revision.

Offline complete-gold controlled index. Recall is only measurable where the relevant set can be enumerated. The controlled index is therefore fully synthetic and self-generated. Each topic has a materialized gold set under the frozen generator, and tests recompute the world and tasks from the recorded seed and compare all committed hashes rather than trusting a hand-authored denominator. Discoverability is heterogeneous: relevant items are partitioned across lexical, semantic, citation, reformulation and restricted/provider routes; two route labels deliberately share a backend; duplicate locators share content identity; revisions share identity but change digest; extraction-question-shift cases require legitimate rereads; and unavailable-route cases distinguish transport censoring from absence. Budgets are below exhaustive probing, so route allocation and stopping remain observable variables.

The completed suite has 390 tasks over five case families, drawn from 78 topics over 1,210 documents, frozen at suite fingerprint `2f6936ba52fb`. Every candidate uses the same host-owned session. The session alone can access the world, meters queries and reads, records route stops and

transport states, and evaluates claims after the candidate returns. The candidate receives only a public task view and label-free retrieved records; protected gold and the relevance rule are never passed through the system interface.

External benchmark tasks. AutoResearchBench provides Wide and Deep scientific-discovery tasks [11]. The access audit establishes a narrower fact than an external result: Wide’s official exact IoU/recall/precision path scores normalized arXiv identifiers without a credential, while its sampled `max_iou_at_k` family is noisy because upstream sampling is unseeded. Deep’s official title-matching evaluator needs an OpenAI-compatible judge. The Wide scorer was exercised on synthetic scorer input to verify that credential independence; no ORION-vs-baseline Wide result is archived in this revision, and the scorer is not claimed to be vendored here.

SAGE is retained as nearest-work and as a lexical-baseline warning [3], but is not executed as if its published setting were available: the audited repository lacks the 200,000-paper retrieval corpus described by the paper and lacks an official evaluator. Any substitute corpus or weighting rule would be a declared deviation rather than the SAGE benchmark. MetaSyn’s separation of retrieval from screening is adopted as a measurement constraint [10]; its retrieval/screening scoring path is credential-free by source audit, but no end-to-end result is archived here. AgentSLR remains a protocol-driven systematic-review neighbor/baseline family rather than an official evaluator dependency [6].

Frozen external retrieval snapshots and live-provider campaign. The prospective external design freezes a gold-blind candidate pool per query before comparing systems, then runs a smaller predeclared live subsample. Pool coverage must be reported as a shared ceiling because a gold-blind pool can omit gold. Live requests are designed to retain URL/parameters, query derivation, timestamp, raw response and typed transport outcome. The capture machinery and manifest exist and have been exercised, but the final result-bearing external pool and live campaign have not been executed in this revision. Therefore no external cost, latency or recall result is inferred from the capture audit.

6.3 Systems under test

The frozen offline companion compares full ORION with six confirmatory system identifiers: `bm25_keyword`, `dense_retrieval`, `sparse_dense_hybrid`, `one_pass_rag`, `agentic_single_route`, and `protocol_driven_systematic_review`. A seventh comparator, `adaptive_multiroute_exploratory`, was kept explicitly exploratory and was not retrofitted into the frozen confirmatory baseline set.

These controlled-index systems are mechanism comparators, not claims to replicate a production BM25 engine, a particular embedding model or an external agent implementation. They consume the public synthetic route interface under identical budgets so that the variable under test is route/read/stopping governance. Their relevance screen sees only the public question and retrieved bibliographic text. The authored synthetic text makes screening deliberately easy; the experiment is not presented as evidence about classifier quality.

Six frozen ablations remove exactly one governance mechanism or authorize one forbidden transition: route-independence checking, the question coordinate in the read ledger, route/task stop separation, unavailable-route open state, coverage-diagnostic non-authority, and content-identity deduplication. All use the same session and budgets as full ORION.

6.4 Identity, reads and stopping

A read is keyed by content identity, content digest and extraction question in the full system. A new question can therefore reopen unchanged content for a legitimate reread, while mirrors with the same content identity/digest do not become new evidence. The corresponding ablations remove the question coordinate or replace content identity with document locator to expose suppression or duplicate work.

Route stop and task closure are separate actions. A route may stop after its public probe space is exhausted or after a typed transport outcome, but task closure is invalid if relevant material remains reachable within budget. An unavailable route remains censored/open rather than being converted to an empty result. Coverage diagnostics may inform a route policy but have no authority to certify global completeness.

6.5 Measurements

The offline evaluator reports denominator-valid complete-gold recall, precision, missed-gold count, route use, route/content overlap diagnostics, marginal relevant gain after the first route, premature task closure, censoring, duplicate processing and legitimate rereads, plus host-owned resource counters. Claims are normalized by content identity rather than locator so mirrors cannot inflate recall. Rich per-run artifacts retain route calls, reads, stop events, authority flags and raw-artifact hashes; the publication summary binds the full 16,380-record and rich-artifact sets by SHA-256.

The external measurement plan additionally defines official Wide/Deep scores, screening metrics and ranking-only metrics where their denominators and task semantics support them. Those definitions do not imply that an external result exists.

6.6 Statistics and authority

The statistical unit is the task. Repeated stochastic runs nest within task and system and never count as additional tasks. The frozen plan defines paired bootstrap comparisons, Wilson intervals for standalone binary rates, superiority/non-inferiority margins, a cost ceiling and Holm correction for the external confirmatory analysis. It also defines an authority ladder. The completed offline suite has $n = 390$, which reaches the plan’s committed tier `TIER_B_committed` ($n = 385$) at an achieved half-width of 0.0496; because that half-width exceeds the frozen superiority margin, the primary offline result carries the plan’s mandatory underpowered label. The three deterministic repeat seeds verify stable execution; they do not manufacture additional tasks. We therefore report descriptive absolute differences and mechanism diagnostics, with no promoted superiority decision from the offline companion.

For a future external confirmatory result, the frozen primary rule requires the family superiority margin of 0.03 under matched resources with the premature closure safety guard no worse than 0.02. Non-inferiority uses the predeclared 0.01 margin and is inadmissible above the 1.25 consumed-cost ratio. Secondary families use Holm correction. These rules remain prospective because no external system result is reported here.

6.7 Failures, exclusions and contamination

Candidate failures, malformed outputs, budget exhaustion, abstention and typed transport failures remain outcomes. Provider unavailability cannot become evidence of absence. The frozen evaluator uses an explicit failure taxonomy and terminal precedence; the publication table therefore warns that a lower-priority mechanism can be masked by a higher-priority terminal

classification such as premature closure. Public benchmark questions and gold identifiers also create a contamination risk for future live search. A credential-free search-time self-exposure diagnostic over the frozen 400-task AutoResearchBench Wide candidate run is archived as `evidence/external_results/AUTORESEARCHBENCH_SEARCH_CONTAMINATION_V1.json` (task rate 0.0). That diagnostic does not claim absence of indirect, training-time, or live-provider contamination.

6.8 Decision rule

The paper-level rule remains fail-closed. The offline companion can support a mechanism claim only at descriptive authority. External discovery superiority requires an archived denominator-valid external result under the frozen matched resources and safety rule. If that evidence is absent, blocked or null, the claim is narrowed rather than promoted by analogy from the synthetic world.

7 Results

7.1 Authority of the result

Two result-bearing evidence families are complete in this revision, with different authority. First, the frozen offline complete-gold companion contains 390 tasks, 14 systems (full ORION, six confirmatory baselines, one explicitly exploratory comparator, and six ablations), and three predeclared repeat seeds, for 16,380 normalized run records. The systems are deterministic; the repeats therefore test harness stability and do not increase the statistical unit beyond 390 tasks. The frozen statistical plan’s committed tier `TIER_B_committed` requires 385 tasks, which the assembled 390 tasks reach at an achieved half-width of 0.0496 for a standalone rate. Because that half-width still exceeds the frozen superiority margin of 0.03, the plan’s underpowered label attaches to the primary offline result: no superiority decision from the offline companion is promoted to a paper-level claim.

Second, the credential-free MetaSyn probe is a bounded external retrieval/screening result. It completed all 86 released test reviews using the pinned MetaSyn BM25 title+abstract retriever followed by the predeclared public-protocol screening rule, and was scored by the pinned official ID-only evaluator. That probe is not the complete multi-provider ORION system and is therefore reported separately from the paper’s still-open matched external ORION-vs-baseline claim.

The offline publication summary is `evidence/offline_results/RESULTS_SUMMARY_V1.json`. It binds the full 16,380-record set by SHA-256 `27b8e55b68a65906fe0971ed2f24a814d31d66a1423d38782bb121c7f06e1525` and the corresponding rich-artifact hash list by `ed5cf7bd823fcabd6a57658a05560bd5f1e30256aa3f8eb8404325c493ed3cfe`. The exact run manifest was committed before the first outcome snapshot. The mechanism projection used for Figures 4–5 is archived separately as `evidence/offline_results/OFFLINE_MECHANISMS_V1.json`; it records the same two source digests and is regenerated from host-owned traces rather than copied from manuscript numbers. The MetaSyn result is bound by `evidence/external_results/METASYN_ID_ONLY_PROBE_V1.json` and its exact integer error decomposition by `METASYN_SCREENING_FN_LEDGER_V1.json`.

7.2 Controlled-index comparison

Full ORION attains mean complete-gold recall 0.979487 and precision 1.0. It passes 319 of 390 tasks, returns `FAIL` on 59 tasks after honestly exhausting its route-call budgets without certifying completeness, and returns `CANNOT_CHECK` on 12 tasks because a route holding unretrieved relevant

System	Recall	Premature closure	Pass	Cannot check	Fail
ORION full	0.979487	0.00	0.82	0.03	0.15
Protocol-driven SLR	0.666667	1.00	0.00	0.00	1.00
Sparse+dense hybrid	0.555556	1.00	0.00	0.00	1.00
One-pass RAG	0.555556	1.00	0.00	0.00	1.00
BM25/keyword	0.333333	1.00	0.00	0.00	1.00
Dense-route comparator	0.333333	1.00	0.00	0.00	1.00
Agentic single route	0.333333	1.00	0.00	0.00	1.00

Table 1: Controlled-index descriptive comparison. Rates are over 390 tasks; **FAIL** on ORION’s row is honest budget exhaustion, not premature closure. The frozen plan’s underpowered label applies to the primary offline comparison, so no superiority interval or decision is reported here. A prospective external Wide campaign was executed but failed its frozen transport validity gate; it is retained separately as **CANNOT_CHECK**.

material becomes unavailable; it never certifies a censored task complete. Its premature-task-closure rate is 0. The strongest of the six protocol-frozen confirmatory baselines is the protocol-driven systematic-review comparator, with mean recall 0.666667. That comparator claims task completion on all 390 tasks while reachable gold remains, so its premature-closure rate is 1.0. The descriptive recall difference is therefore +0.31282, with a descriptive premature-closure difference of -1.0. These numbers are evidence about the authored controlled world, not an estimate of real-web superiority.

The two systems are identical through their first three search calls in this authored world. The protocol-driven comparator then reaches mean recall 0.666667 by query 7 and remains at that value through query 12. ORION is at 0.757835 by query 7 and continues to gain, reaching 0.979487 by query 12. Figure 3 therefore localizes the descriptive gap to evidence gained after the early common search prefix rather than to an initial retrieval advantage.

7.3 Mechanism ablations

The safety ablations fail in the directions the frozen design intended them to probe. Removing the route-independence check reduces mean recall to 0.444444 and produces premature closure on every task. Allowing a route stop to certify task closure yields recall 0.333333 with the same 100% premature-closure rate. Letting a coverage diagnostic acquire stopping authority yields recall 0.555556 and again closes every task prematurely. These are negative controls: their purpose is to demonstrate that the forbidden transitions are observable, not to serve as competitive systems.

Removing content-identity deduplication is less catastrophic but measurably wasteful in the controlled world: mean duplicate-processing rate rises from 0 to 0.101446, 145 of 390 tasks exhaust the read budget, pass rate falls from 0.817949 to 0.605128, and mean recall falls to 0.952707. Removing the question coordinate from the read ledger leaves aggregate recall unchanged at 0.979487 but changes the extraction-question-shift mechanism exactly where expected: the mean count of legitimate rereads on that family falls from 3.666667 to 1.0. This is a mechanism result, not evidence of a recall gain from the ledger. Finally, removing the unavailable-route open state leaves mean recall numerically unchanged but turns the 12 censored **CANNOT_CHECK** cases into asserted completions and premature-closure failures. The distinction is therefore one of epistemic safety rather than headline recall.

The lexical route contributes 3.0 first relevant identities per task on average. Semantic and citation routes each contribute 2.0 additional identities, reformulation contributes 0.848718, and the restricted route contributes 0.966667; the last mean is below one because tasks that end censored

or budget-exhausted before the restricted route is consulted contribute zero. Contribution and independence are deliberately separate variables: reformulation adds relevant content but shares the lexical backend and therefore cannot be promoted to an independent capture occasion.

Figure 5 exposes why ORION does not equate numerical disjointness with route independence. In particular, lexical and reformulation have zero mean content overlap here yet are still rejected as independent because they share a backend. Conversely, lexical and semantic have nonzero mean overlap (0.07557) but remain structurally independent because neither backend nor query derivation is shared. The independence decision is therefore made from capture construction, while overlap remains a measured diagnostic.

7.4 MetaSyn external retrieval and screening probe

The corrected MetaSyn probe completed all 86 released test reviews with no failed review. Under the pinned official ID-only evaluator, macro retrieval recall is 0.7485 and retrieval precision 0.0627. Conditional retention among retrieved reference items is 0.7224; final inclusion recall is 0.5651, inclusion precision 0.1426 and inclusion F1 0.2033. The evaluator reports post-retrieval loss 0.1834 and screening accuracy 0.7228. Mean candidate output is fixed at 200 retrieved and 60 included items per review by the declared probe budget.

The evaluator-side integer ledger makes the stage loss concrete. Across the 86 reviews there are 1,677 reference-article links. The top-200 retrieval pools contain 1,079 of those links, so 598 are retrieval false negatives. Of the 1,079 retrieved reference links, 736 survive the deterministic screen and 343 are lost post retrieval. The final included sets therefore miss 941 reference links in total. Gold is loaded only on the evaluator side after candidate output is frozen, and fresh regeneration verifies every deterministic official per-review metric before emitting the counts. These results identify where the bounded probe fails; they do not establish superiority for full ORION.

7.5 Failure analysis and external status

Under the evaluator’s fixed terminal-failure precedence, premature closure is the dominant failure class in the controlled companion; the exploratory adaptive comparator also exposes 65 transport-failure classifications, while the no-dedup ablation exposes 145 budget-exhaustion failures. Full ORION’s only terminal failure classification is honest budget exhaustion on 59 tasks: 319 tasks pass, 12 are censored, and premature closure remains at zero. The full per-system taxonomy is archived beside the publication summary as Table P2-3.

The gold-defined O1 route-stop replay is archived as supplementary Table P2-S1. Full ORION makes 13 route-stop false positives in 1,950 route-stop events (0.006667): those restricted-route cases still have a theoretically reachable gold identity when the route stops. These do not become task-level false closures. The O4 unavailable-route obligation remains open, so ORION returns `CANNOT.CHECK` rather than asserting task completeness. Among the 1,878 ORION task-routes that reach the O1 oracle exhaustion point, there are zero route-stop false negatives. The restricted route records 377 post-exhaustion attempts in total; no task takes the two or more further attempts required for an O1 false negative. This null-on-FN result is reported rather than being silently omitted, and it demonstrates why route-stop error and task-stop safety must be measured separately.

The remaining external authorities are kept separate from the completed MetaSyn probe. AutoResearchBench Wide’s released scorer is deterministic and credential-free, while the official Deep metric uses the released LLM title judge. That judge has now executed for the declared keyless public-arXiv probe: all 600 released Deep tasks were scored under the pinned judge prompt and

model reached through a local protocol adapter, and the judge matched the reference title on 0 of them, a hit rate of 0.000. A nine-case positive/negative judge control—verbatim, truncated, and case-variant matches alongside distinct-paper non-matches—passed in full before the archive was accepted, so a zero-match outcome is attributable to the probe rather than to an always-no-match judge failure. This is a bounded external probe, not a matched ORION-versus-baseline Deep comparison, and no Deep superiority claim is drawn from it. SAGE still lacks both the published retrieval corpus and an official evaluator. The frozen three-route live campaign also remains `CANNOT_CHECK`: its arXiv route is credential-free, but its two OpenAlex-backed route families require a funded `OPENALEX_API_KEY`, which the verified same-repository preflight did not find. No absent credential is converted into a negative outcome or a zero-cost measurement.

A separate gold-blind OpenAIRE identity discriminator first established that structured `scheme=arxiv` PID objects were available on 3 of 12 frozen public Wide questions, with 12 of 12 HTTP requests successful; that discriminator was forbidden to score or promote the paper. A subsequently frozen 400-row OpenAIRE/Crossref matched campaign then issued exactly 1,200 logical provider requests over one shared capture. The 400 Crossref discovery calls and 400 direct OpenAIRE calls succeeded, but every predeclared OpenAIRE DOI-crosswalk call returned HTTP 400. The observed provider-success fraction was therefore 0.666667, below the frozen 0.90 validity gate. Without a valid cross-backend route, the balanced baseline, direct-first diagnostic and governed projection were byte-identical; their paired IoU differences were zero on all 399 distinct questions. The frozen analyzer correctly emitted `P2_WIDE_EXTERNAL_CANNOT_CHECK`, not a valid negative result. The terminal, workflow receipt and expiring Actions archives are retained as `P2_WIDE_OPENAIRE_MATCHED_RESULT_V1.json`, `P2_WIDE_OPENAIRE_MATCHED_RUN_RECEIPT_V1.json`, and checksummed mirrors under `evidence/ci_mirror/`. A new campaign would require a new prospective freeze after independently validating the documented structured crosswalk request syntax; this failed capture cannot be reinterpreted after the fact.

Therefore the externally supported full-ORION discovery-benefit claim remains `CANNOT_CHECK`. The controlled-index result supports the narrower statement that the frozen governance mechanisms behave as designed under a complete denominator and matched host-owned budgets; the MetaSyn result supports only its declared external retrieval/screening probe. Bounded AutoResearchBench probe results, when archived, must remain separately labeled unless they exercise the final matched ORION system under the frozen promotion plan.

8 Nearest work

AutoResearchBench shows that both specific-target and open-ended scientific literature discovery remain difficult [11]. SAGE demonstrates that strong lexical retrieval can outperform LLM retrievers inside deep-research workflows [3], while AgentIR shows that an agent’s reasoning trace can itself become a retrieval signal [1]. SIEVE further demonstrates that fielded Boolean constraints, structure-rich inspection and selective section fetching can improve deep-research retrieval efficiency [9]. Paper II therefore does not claim lexical, reasoning-aware, field-aware or generic adaptive retrieval as novel; these are candidate acquisition modules whose outputs remain subject to the same closure authority boundary.

MetaSyn separates retrieval from eligibility/screening and shows why retrieval ceilings do not imply inclusion recall [10]. AgentSLR demonstrates that end-to-end systematic-review automation is itself an active baseline area [6]. Bibliography expansion can also transform literature-search recall, while human reference lists are not a complete single-axis ground truth [8]. These results motivate stage attribution, donor composition and denominator caution rather than a claim that

ORION invented literature-search automation.

The stopping boundary is also crowded prior art. DeepControl regulates retrieval continuation using marginal information utility [12]; decision-theoretic screening rules optimize stopping against explicit cost/payoff models [2]; HALT frames stopping as claim-evidence coverage and can abstain when coverage is not verifiable [7]; MiCP applies conformal prediction to adaptive multi-turn reasoning to maintain target coverage while permitting early stopping [13]; and a 2026-08-13 Search-R1 study trains structured sufficiency/gap STOP judgments while explicitly not claiming safe stopping [4]. Accordingly, Paper II does not claim verification-aware, conformal-coverage, utility-based or learned STOP/CONTINUE policies as novel. Its candidate residual contribution is an authority rule: relevance, utility, coverage and sufficiency may guide acquisition or route-local stopping, but unresolved unavailable/censored route obligations cannot be silently erased or promoted into global scientific closure.

Question-conditioned memory is likewise broader than this paper: MemChain constructs question-conditioned evidence plans and grounded memory traces [5]. The residual here is only the dual bookkeeping needed for literature discovery—content identity to prevent duplicate discovery inflation, kept distinct from question-conditioned processing state so a changed extraction question can legitimately trigger rereading.

Because directly relevant stopping work appeared as recently as 2026-08-13, the nearest-work state is frozen only through 2026-08-17 and is not labelled saturated. The source-bound donor ledger and pre-submission refresh rule treat later material work as a reopen trigger rather than silently preserving stale novelty.

9 Limitations and integrity

The completed controlled experiment is deliberately synthetic. Its complete denominator is what makes missed relevant material and premature closure observable, but the authored routes, public probe vocabulary and easy screening text make it a mechanism test rather than a realistic estimate of retrieval difficulty. The observed +0.31282 recall difference therefore cannot be transferred to the open web or to AutoResearchBench by analogy.

The suite carries 390 tasks, which reaches precision tier TIER_B_committed at an achieved half-width of 0.0496 for a standalone rate. The frozen statistics plan fixed those tiers before outcomes were inspected and assigns this campaign authority TIER_B_committed; because the achieved half-width still exceeds the frozen superiority margin of 0.03, the primary offline result carries the plan’s underpowered label whatever the task count becomes. All 16,380 system/task/repeat executions reproduce exactly, and the 3 deterministic repeat seeds demonstrate stable execution; repeats nest within task and never increase the statistical unit.

Open-world completeness remains generally unprovable from finite search. External providers are mutable, rate limited and differently scoped, and public benchmark questions/gold identifiers create search-time contamination risk. The corrected MetaSyn run is external but narrow: it uses pinned BM25 retrieval plus a deterministic public-protocol screening rule and the official ID-only evaluator, not the complete ORION route system. Its exact evaluator-side ledger contains 1,677 reference links, of which 598 are missed at retrieval and another 343 are lost after retrieval, leaving 941 final false negatives. These numbers characterize that probe and are not promoted to a general ORION claim.

Several external authorities remain partially unavailable rather than negative. AutoResearchBench Deep official scoring has now executed through a local OpenAI-compatible adapter for the bounded keyless probe, with the judge control passing in full, but the matched multi-provider Deep

comparison remains unexecuted. SAGE still lacks its published retrieval corpus and official evaluator. The frozen three-route live-provider protocol requires a funded OpenAlex key for two of its three route families; a same-repository preflight found no `OPENALEX_API_KEY`, so the protocol cannot honestly enter `EXECUTION_FROZEN` here. Provider or credential unavailability is therefore retained as `CANNOT_CHECK`, never converted into evidence of absence or a flattering zero-cost result.

The mechanism ablations themselves impose an interpretive limit. Removing the question coordinate from the read ledger changes legitimate reread behavior but does not change aggregate recall in this suite; removing the unavailable-route open state changes censoring/closure semantics while leaving aggregate recall numerically unchanged. These null-on-recall mechanism findings are retained rather than converted into a stronger claim.

All result-bearing statements in the abstract, Results and conclusion are mapped to immutable artifacts in `evidence/CLAIM_LEDGER_V1.md`. Rows without an archived supporting result remain `CANNOT_CHECK`; an access audit, implementation test, or credential preflight cannot substitute for a system outcome.

10 Data and code availability

The ORION repository contains the controlled index, host-owned evaluation harness, the frozen offline systems and ablations, the descriptive analysis code, and the publication-layer regeneration commands. The exact subject revision and Git blob identities used for the result-bearing offline campaign are recorded in `protocol/OFFLINE_RUN_MANIFEST_V1.json`. The frozen complete-gold world is distributed under `evidence/offline_gold/`; it is fully synthetic, regenerable from the recorded seed, and verified on every clean CI run against its committed content hashes and suite fingerprint.

The compact publication result is `evidence/offline_results/RESULTS_SUMMARY_V1.json`. The command `python3 papers/paper-02-open-world-scientific-discovery/scripts/run_offline_companion.py --check` reconstructs all 16,380 runs and refuses drift. With `--write-raw DIR` it also emits the complete normalized JSONL result set and all rich run artifacts. The committed summary records SHA-256 digests of both complete sets, so a reproduction can be checked without requiring hundreds of generated files in the repository.

External benchmark artifacts are not redistributed. Their licences, access states and pinned revisions are recorded per artifact in `protocol/EXTERNAL_ACCESS_AUDIT_V1.json`, with source checks retained under `evidence/access/` and summarized in Table P2-1. AutoResearchBench is pinned to upstream commit `a46c9bfb8968786f73f0a6a5b365b5384cd0f96d`; the released 1,000-record bundle is split by its explicit task-type field into 600 Deep and 400 Wide records before candidate custody. The Wide scorer is deterministic and credential-free. Deep official scoring uses the released title-identity judge and still requires an OpenAI-compatible endpoint; the repository workflow now records this authority precondition without exposing credential values.

The completed MetaSyn probe is archived as `evidence/external_results/METASYN_ID_0_NLY_PROBE_V1.json`. It binds all 86 released test reviews, the pinned upstream revision, dataset revision, candidate-output tree and official ID-only evaluator output. The evaluator-side integer error ledger is separately archived as `METASYN_SCREENING_FN_LEDGER_V1.json`; its regeneration reloads the pinned gold only after candidate output is frozen and verifies the official per-review metrics before emitting exact retrieval and screening false-negative counts. These artifacts make the bounded MetaSyn retrieval/screening result reproducible without reclassifying it as full ORION.

The failed OpenAIRE/Crossref Wide campaign is preserved rather than left behind an expiring Actions URL. The exact result and run receipt are stored under `evidence/external_results/`;

checksummed mirrors of the shared public candidate capture, failed first evaluator handoff, and successful evaluator-only repair are under `evidence/ci_mirror/`. The mirror manifest records both archive-level and per-file SHA-256 digests. Evaluator gold never entered the candidate capture. The campaign remains `CANNOT_CHECK` because 400 of 1,200 planned provider calls failed, not because the evidence was discarded.

The live-provider capture implementation and campaign manifest retain request URL/parameters, query derivation, timestamp, raw body, hash and typed transport status. The frozen live protocol requires arXiv plus two OpenAlex-backed route families. A non-leaking Actions preflight is archived as `evidence/access/LIVE_PROVIDER_PREFLIGHT_2026-08-16.json`; it confirms that arXiv is credential-free but that no funded `OPENALEX_API_KEY` is available in the same-repository execution context. Therefore the three-route live campaign cannot enter `EXECUTION_FROZEN` here, and no raw live campaign or live monetary-cost ledger is fabricated. Operator re-scope 2026-08-17 removed the empirical-cost requirement: the sanctioned cost axis is the host-recorded search-query count (Figure 3, Table P2-2), not a live OpenAlex or arXiv API bill.

11 Reproducibility

Four levels are distinguished, and we claim only what the current artifacts support.

Fully reproducible without network access. The 390-task offline controlled-index companion regenerates from committed data and subject code. A clean GitHub Actions job independently rebuilds the 16,380-run publication projection and compares it to the committed summary. The frozen plan awards the result its committed `TIER_B_committed` authority together with a mandatory underpowered label on the primary offline comparison; reproduction does not elevate that authority.

Reproducible bounded external evidence. The pinned MetaSyn workflow reconstructs the public/gold split, runs the credential-free BM25 plus public-protocol screening candidate, executes the official ID-only evaluator, and can regenerate the exact false-negative ledger. This is external evidence for that declared probe only.

Runnable credential-free benchmark lanes. AutoResearchBench Wide scoring is deterministic once the gold-blind candidate output is frozen. The same public-arXiv candidate machinery also supports the declared deterministic Deep exact-ID deviation. The official Deep title judge remains a distinct authority and is not replaced by the ID deviation when no judge endpoint is available.

Executed official Deep judge archive. The bounded keyless public-arXiv probe of the official Deep LLM title judge is archived as `evidence/external_results/DEEP_OFFICIAL_ARCHIVE_V1.json` with its nine-case positive/negative judge control in `DEEP_JUDGE_CONTROL_2026-08-17.json`. The checkpointed judge runner and the archive finalizer are both reproducible from the repository as `scripts/judge_checkpointed.py` followed by `scripts/finalize_deep_scoring.py`; the archive records the judge model, the pinned prompt digest, the run manifest hashes, and the bounded probe scope label.

Capture-capable but blocked as a final live campaign. The committed live layer is designed to archive mutable provider evidence rather than re-derive it, but the frozen route set requires funded OpenAlex access that is absent in the verified execution context. Until that authority is supplied and the campaign is actually run, the paper keeps raw live-provider outcomes,

empirical live monetary cost/wall-clock and the matched external discovery claim at `CANNOT_CHECK`. Query-count metrics from the offline companion are the completed cost evidence in this revision.

12 Ethics

This work does not collect new human-subject records or personally identifiable information. The frozen complete-gold companion is fully synthetic and regenerable from a recorded seed. External probes use publicly released benchmark questions and evaluator interfaces; third-party corpora are not redistributed where Table P2-1 records a licence block or partial redistribution. Dual-use risk is that of any scientific-literature retrieval system: the paper evaluates discovery governance and does not present a deployment, surveillance, or human-facing study.

13 Conclusion

Paper II advances a systems thesis broader than a contest between individual retrievers: acquisition quality and scientific-closure authority are different layers. The frozen controlled-index companion supports a narrow engineering result: under a complete denominator and matched host-owned budgets, the route/read/stopping governance mechanisms behave in the intended fail-closed directions, and the strongest confirmatory controlled comparator still leaves substantial reachable gold. The separate MetaSyn probe supplies real external retrieval/screening evidence but does not exercise or validate the complete multi-route ORION system. The controlled evidence remains descriptive and synthetic, and the paper’s matched external discovery/stopping claim remains open until denominator-valid Wide/Deep or other admissible external results are archived. If those authorities remain unavailable or do not support the frozen promotion rule, the claim must remain narrowed rather than promoted. This positions the architecture as an authority contract beneath replaceable acquisition modules: lexical, metadata-augmented, reasoning-aware, field-aware, citation-expansion, verification, conformal, sufficiency and utility mechanisms may be composed without granting local signals permission to erase unresolved obligations.

References

- [1] Zijian Chen, Xueguang Ma, Shengyao Zhuang, Jimmy Lin, Akari Asai, and Victor Zhong. Agentir: Reasoning-aware retrieval for deep research agents. *arXiv preprint arXiv:2603.04384*, 2026.
- [2] Aaron H. A. Fletcher and Mark Stevenson. Decision-theoretic stopping rules for document screening. *arXiv preprint arXiv:2606.07071*, 2026.
- [3] Tiansheng Hu, Yilun Zhao, Canyu Zhang, Arman Cohan, and Chen Zhao. Sage: Benchmarking and improving retrieval for deep research agents. *arXiv preprint arXiv:2602.05975*, 2026.
- [4] Weimeng Luo. When should multi-round rag stop? structured stopping judgments and retrieval reduction in search-r1. *arXiv preprint arXiv:2608.13237*, 2026.
- [5] Yiwen Ma, Songjun Tu, Qichao Zhang, Dong Li, Linjing Li, and Dongbin Zhao. Memchain: Learning interpretable memory traces for memory-augmented llm agents. *arXiv preprint arXiv:2607.24097*, 2026.

- [6] Shreyansh Padarha et al. Evaluating ai-based scientific knowledge synthesis with epidemiological systematic reviews. *arXiv preprint arXiv:2603.22327*, 2026. Introduces the AgentSLR evaluation harness and its expert-annotated dataset.
- [7] Daeyoung Roh and Donghee Han. Halt: Verification-aware stopping for retrieval-augmented search agents. *arXiv preprint arXiv:2608.02009*, 2026.
- [8] Gaurav Sahu, Laurent Charlin, and Christopher Pal. Rethinking literature search evaluation: Deep research helps, and human citation lists are not a ground truth. *arXiv preprint arXiv:2605.29234*, 2026.
- [9] Shuai Wang, Haodong Chen, Yu Yin, Shengyao Zhuang, Bevan Koopman, and Guido Zuccon. Search, inspect, fetch: Exploiting boolean retrieval for deep-research agents. *arXiv preprint arXiv:2608.02751*, 2026.
- [10] Anzhe Xie, Weihang Su, Yujia Zhou, Yiqun Liu, Min Zhang, and Qingyao Ai. Metasyn: A benchmark for llm agents on meta-analysis articles from nature portfolio. *arXiv preprint arXiv:2606.17041*, 2026.
- [11] Lei Xiong et al. Autoresearchbench: Benchmarking ai agents on complex scientific literature discovery. *arXiv preprint arXiv:2604.25256*, 2026.
- [12] Siheng Xiong, Oguzhan Gungordu, Blair Johnson, James C. Kerce, and Faramarz Fekri. Scaling search-augmented llm reasoning via adaptive information control. *arXiv preprint arXiv:2602.01672*, 2026.
- [13] Xiaofan Zhou, Huy Nguyen, Bo Yu, Chenxi Liu, and Lu Cheng. Adaptive stopping for multi-turn llm reasoning. *arXiv preprint arXiv:2604.01413*, 2026.

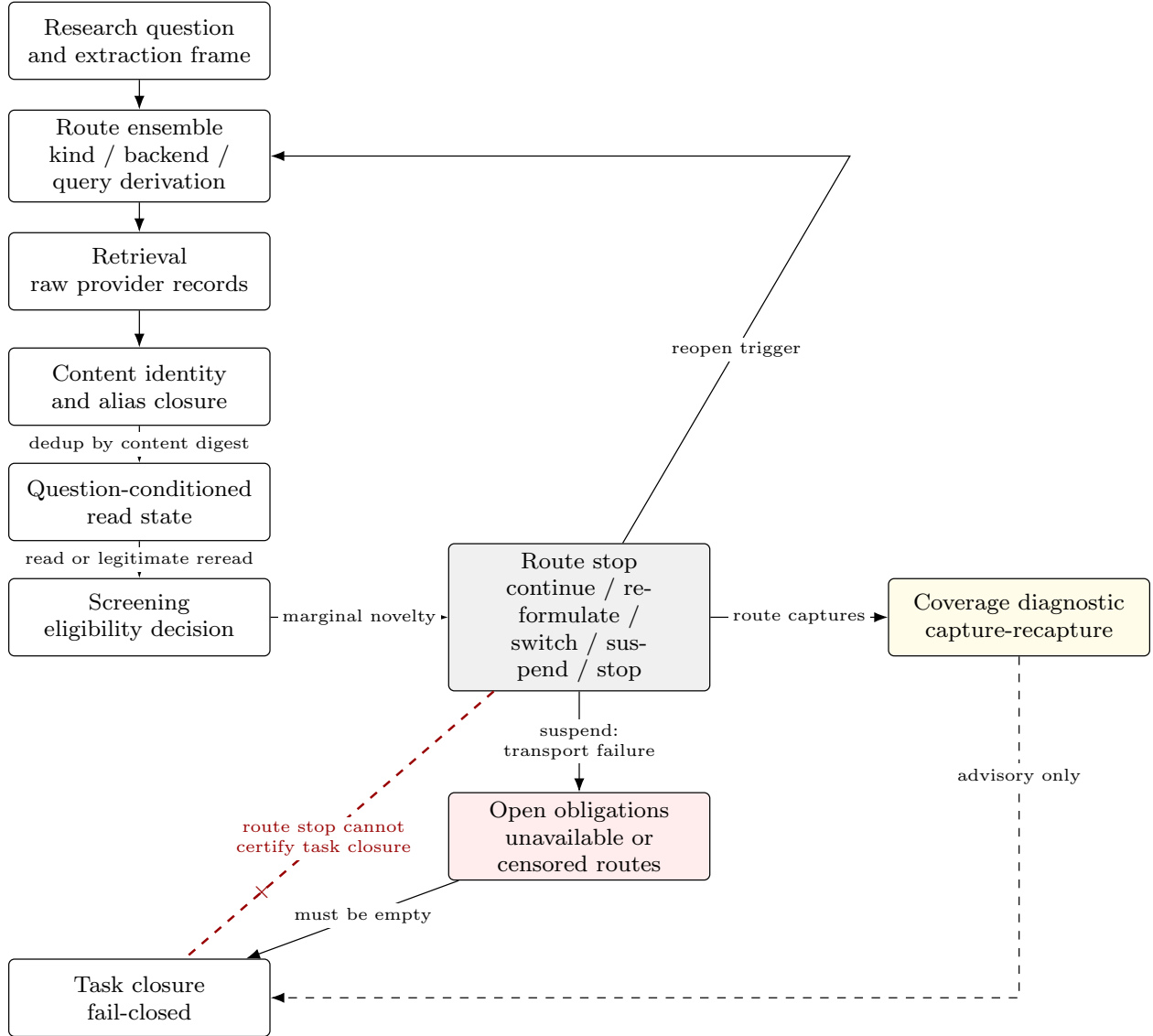


Figure 2: **Figure P2-1.** The discovery pipeline, separating retrieval, screening, route stopping and task closure. Solid arrows are authority-bearing transitions; the dashed arrow marks the coverage diagnostic, which informs but never certifies closure; the crossed arrow marks the transition the design forbids, namely a route stop certifying task-level completeness (Proposition 2). A route suspended by transport failure leaves an open obligation that closure must clear, so provider unavailability cannot become evidence of absence. The figure is generated from the same specification the tests read, so it cannot drift from the state machine it illustrates.

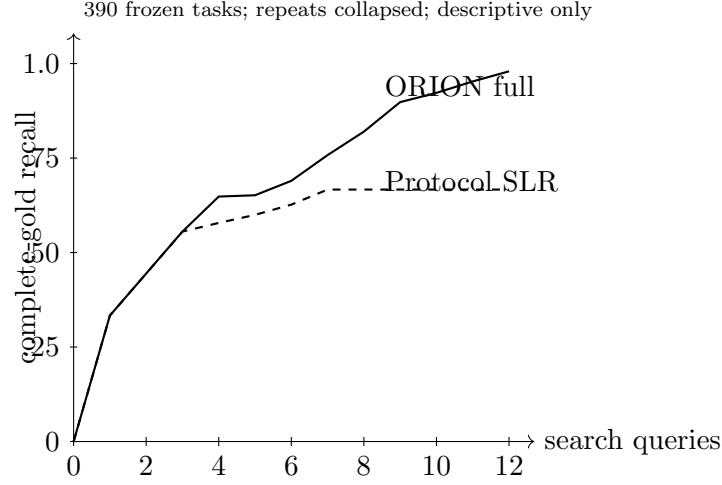


Figure 3: **Figure P2-2.** Cumulative complete-gold recall versus the host-recorded number of search queries for full ORION and the strongest frozen confirmatory baseline. Deterministic repeats are first checked for identical traces and then collapsed within task; the curves are means over the 390 tasks and are descriptive only. Synthetic wall-clock is intentionally fixed to zero, so the controlled companion supports the query axis, not an empirical-time claim.

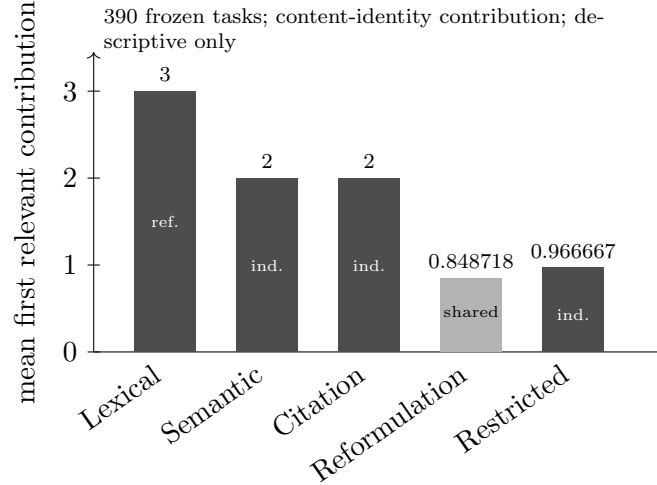


Figure 4: **Figure P2-4.** Mean number of first relevant content identities contributed by each route in full ORION’s campaign order. “Independent” is an earned structural annotation relative to the lexical reference route; the reformulation route is deliberately marked shared-backend even though it adds a new identity. Values are task means over the frozen descriptive companion.

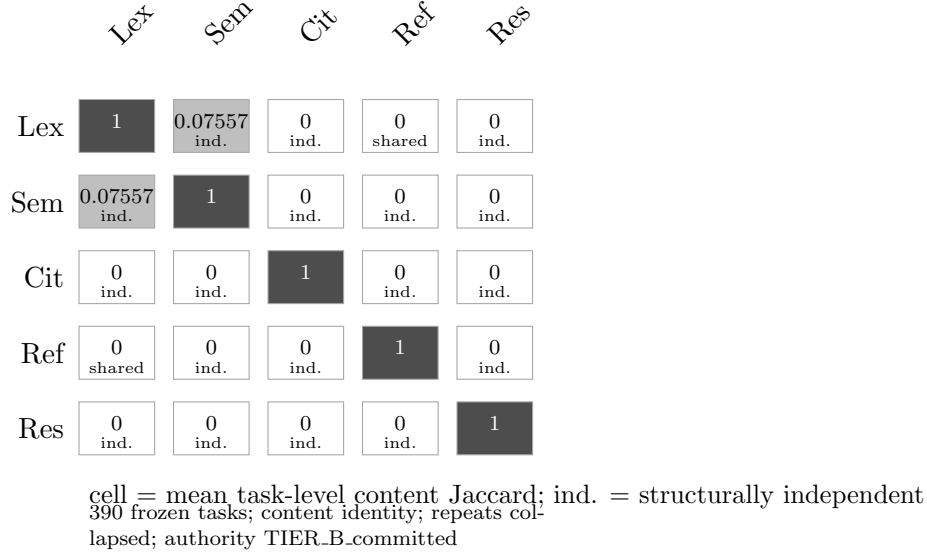


Figure 5: **Figure P2-5.** Mean task-level Jaccard overlap between route captures, computed over content identities rather than source locators, together with the structural independence verdict for each off-diagonal pair. The lexical–semantic overlap is 0.07557; the other off-diagonal mean overlaps are zero in this controlled world. Zero observed overlap is not itself evidence of independence.

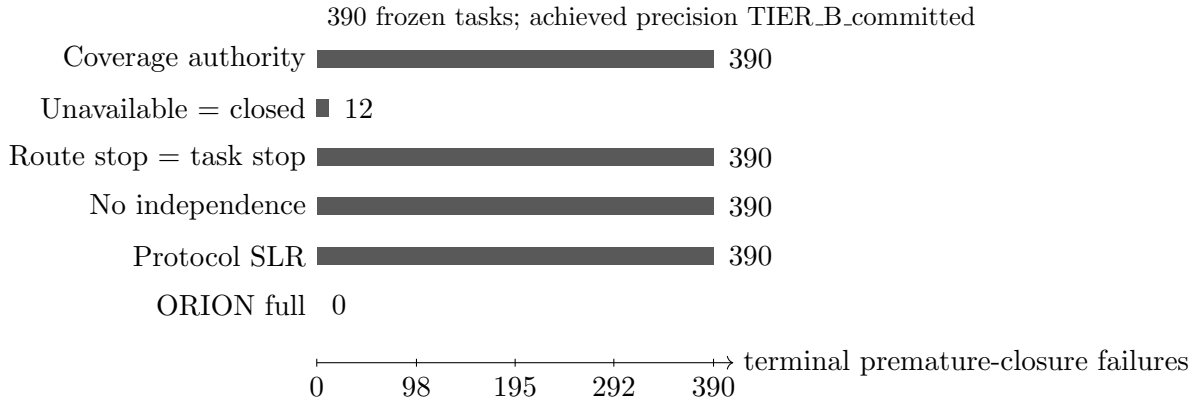


Figure 6: **Figure P2-6.** Terminal premature-closure classifications for selected stopping-safety controls in the frozen 390-task offline companion. The plot is generated from the immutable publication summary. It shows task counts, not an inferential estimate; lower-precedence failure mechanisms can be masked by the evaluator’s terminal-failure precedence.